

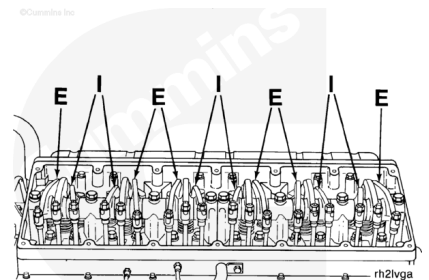
Trainings ▾

Adjust

General Information

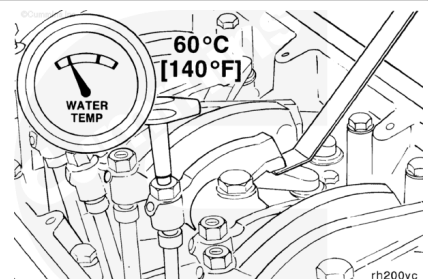
The valves and injectors **must** be correctly adjusted for the engine to operate efficiently. Valve and injector adjustment **must** be done using the values listed in this section. The accompanying table gives the adjustment specifications.

Note : If the valves and injectors have been adjusted during troubleshooting or before the 600 hour scheduled interval, adjustment is **not** required at this time.



Valve Lash Specifications			
	mm		in
Intake Valve	0.36	MIN	0.014
Exhaust Valve	0.69	MIN	0.027

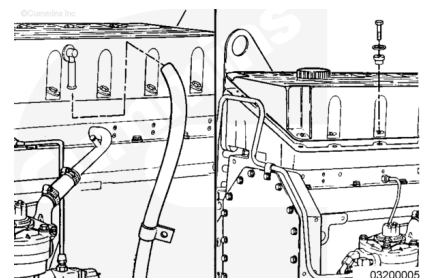
All valve and injector adjustments **must** be made when the engine is cold (stabilized coolant temperature at 60°C [140°F] or below).



Remove the crankcase breather tube from the crankcase breather outlet.

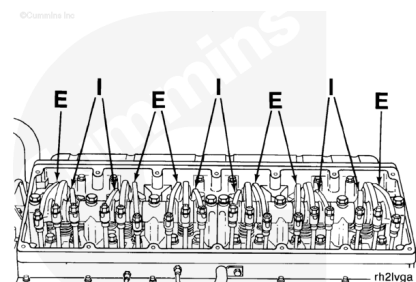
Remove the 16 capscrews, isolators, and spacers from the rocker lever cover.

Remove the rocker lever cover and gasket.



Each cylinder has three rocker levers:

- The long rocker lever (E) is the exhaust lever.
- The center rocker lever is the injector lever.
- The short rocker lever (I) is the intake lever.

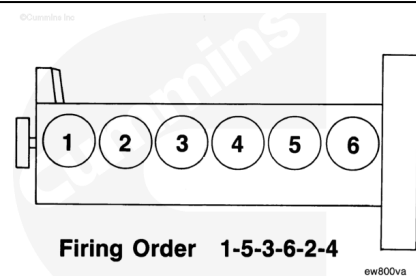


Refer to the accompanying figure for the intake and exhaust valve rocker lever locations.

The cylinders are numbered from the front gear housing end of the engine.

The crankshaft rotation is **clockwise** when viewed from the front of the engine.

The engine firing order is 1-5-3-6-2-4.

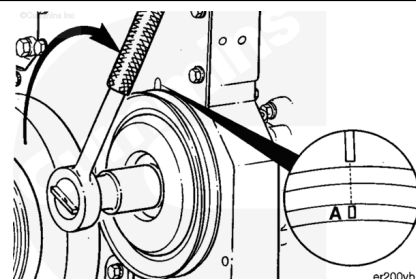


See the example before attempting to begin the adjusting procedure.

The valves and injectors on the same cylinders are adjusted at the same index mark on the accessory drive pulley.

One pair of valves and one injector are adjusted at each pulley index mark before rotating the accessory drive to the next index mark.

Two crankshaft revolutions are required to adjust all the valves and injectors.

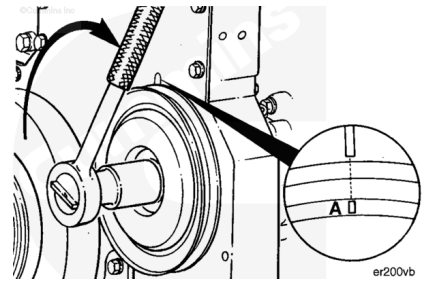


Injector and Valve Measurement Sequence

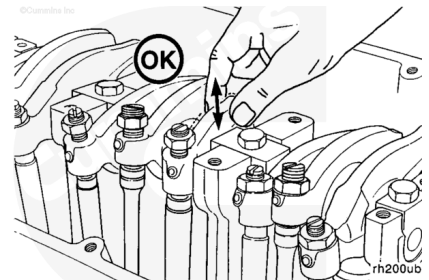
Bar engine in direction of rotation	Pulley position	Set Cylinder	
		Injector	Valve
Start	A	1	1
Advance to	B	5	5
Advance to	C	3	3
Advance to	A	6	6
Advance to	B	2	2
Advance to	C	4	4
Firing Order: 1-5-3-6-2-4			

The adjustment can begin on any valve set mark. In the following example, the adjustment will begin on the "A" valve set mark with cylinder number 1 valves closed and ready for adjustment.

Rotate the accessory drive **clockwise** until the "A" valve set mark on the accessory drive pulley is aligned with the pointer on the gear cover.



When the "A" mark is aligned with the pointer, the intake and exhaust valves for cylinder Number 1 **must** be closed. If these conditions are **not** correct, cylinder number 6 injector and valves **must** be ready to set. Set the injector and valves on the cylinder so that both the intake and exhaust valve rocker lever arms are loose and can be moved from side-to-side.



Injectors

Loosen the injector adjusting screw locknut.

Using a screwdriver and the adjusting screw, bottom the injector plunger three or four times to remove any fuel in the injector.

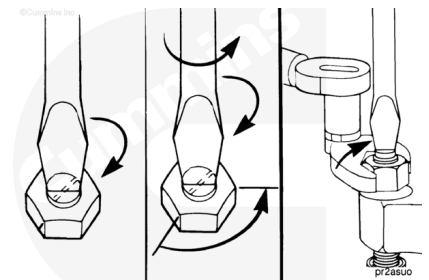
Turn the adjusting screw in until it just bottoms the plunger.

Note : Do not use excessive force when bottoming the plunger.

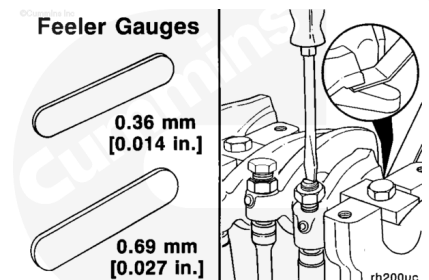
Back out the adjusting screw two flats, 120 degrees.

Hold the adjusting screw, and tighten the locknut.

Torque Value: 61 n•m [45 ft-lb]



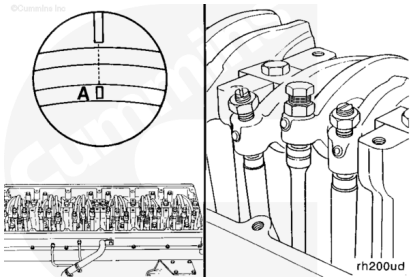
After setting the injector, set the valves on the same cylinder.



Valves

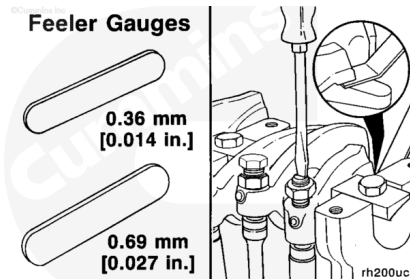
With the "A" valve set mark aligned with the pointer on the gear cover and both valves closed on the cylinder to be adjusted, loosen the adjusting screw locknuts on the intake and exhaust

valves.



Select a feeler gauge for the correct valve lash specification.

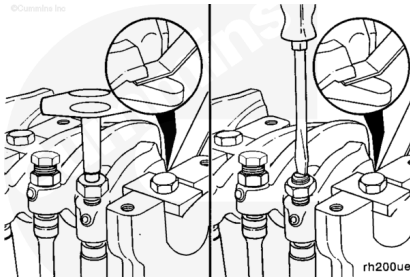
Valve Lash Specifications			
	mm		in
Intake Valve	0.36	MIN	0.014
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Insert the feeler gauge between the top of the crosshead and the rocker lever pad.

Two different methods for establishing valve lash clearance are described below. Either method can be used; however, the torque wrench method has proven to be the most consistent. It eliminates the need to feel the drag on the feeler gauge.

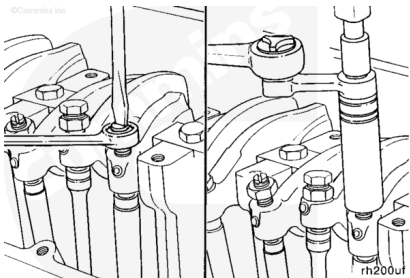
- Torque Wrench Method: Insert the correct feeler gauge. Use an inch-pound torque wrench, Part Number 3376592, normally used to set preload on top-stop injectors, and tighten the adjusting screw.



Torque Value: 0.7 n•m [6 in-lb]

- Touch Method: Tighten the adjusting screw until a slight drag is felt on the feeler gauge.

Hold the adjusting screw in the position shown. The adjusting screw **must not** turn when the locknut is tightened.



Torque Value:

Without torque wrench adapter:

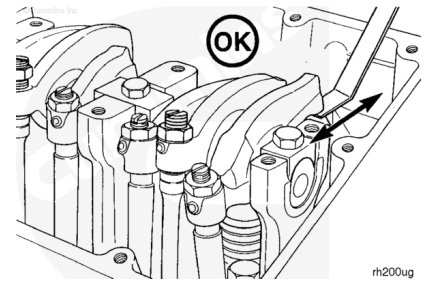
1. 61 n•m [45 ft-lb]

Torque Value:

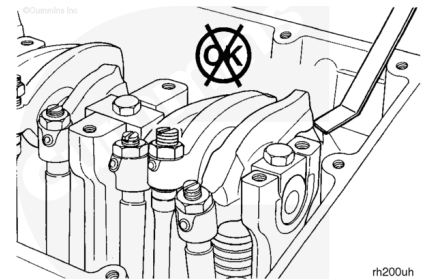
With torque wrench adapter, Part Number 3163196:

1. 47 n•m [35 ft-lb]

After tightening the locknut to the correct torque value, check to make sure the feeler gauge will slide backward and forward between the crosshead and the rocker lever with **only** a slight drag.



If using the touch method, attempt to insert a feeler gauge that is 0.03 mm [0.001 in] thicker between the crosshead and the rocker lever pad. The valve lash is **not** correct when a thicker feeler gauge will fit.



Install the cover on the rocker lever housing.

Install the 16 isolators, spacers, and capscrews in the cover.

Tighten the capscrews in the sequence shown.

Torque Value: 15 n•m [130 in-lb]

